

Answer **all** questions in the answer booklet provided. Please start each question on a new page. Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

1. [Maximum mark: 26]

This question asks you to investigate lines normal to curves of the form $y = \frac{k^2}{x}$.

The curve H has equation $y = \frac{1}{x}$ where $x \in \mathbb{R}, x \neq 0$.

(a) A line N is normal to H at $x = t$.

(i) Show that the gradient of N is t^2 . [2]

(ii) Hence, show that the equation of N is $y = t^2x + \frac{1}{t} - t^3$. [1]

(b) The equation for N given in part (a)(ii) is of the form $y = mx + c$.

(i) Show that either $c = \frac{1}{\sqrt{m}}(1 - m^2)$ or $c = \frac{1}{\sqrt{m}}(m^2 - 1)$. [4]

(ii) Determine the set of values of m for which there exists at least one line normal to H . [1]

(c) Hence, or otherwise, determine the set of values of m for which there exists exactly

(i) one line normal to H ; [1]

(ii) two lines normal to H . [2]

(d) On the same set of axes, sketch

(i) the curve H ; [1]

(ii) for an appropriate value of m , two lines that satisfy the result found in part (c)(ii). Clearly indicate the point at which each line is normal to H . [2]

You are **not** required to state the equations of these lines nor determine where they intersect H or the coordinate axes.

(This question continues on the following page)

(Question 1 continued)

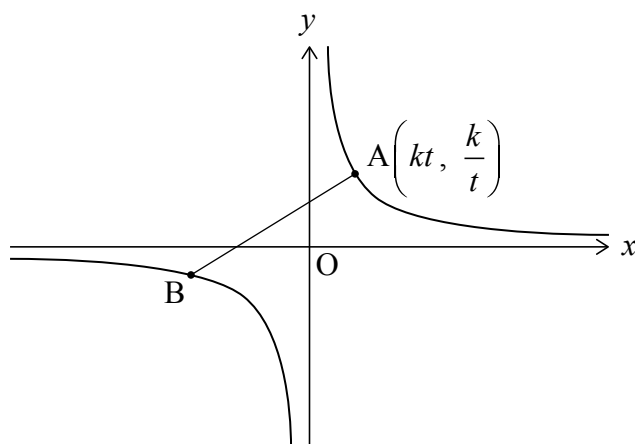
The curve F has equation $y = \frac{k^2}{x}$ where $x \in \mathbb{R}, x \neq 0$ and $k \in \mathbb{R}, k \neq 0$.

The point $A\left(kt, \frac{k}{t}\right)$, where $t \in \mathbb{R}, t \neq \pm 1$, lies on F .

The line normal to F at A intersects F again at point B .

The line segment $[AB]$ is shown in the following diagram.

diagram not to scale



(e) The equation of the line normal to F at A is given by $y = t^2x - kt^3 + \frac{k}{t}$.

(i) Show that the x -coordinates of A and B satisfy the quadratic equation

$$x^2 - k\left(t - \frac{1}{t^3}\right)x - \frac{k^2}{t^2} = 0. \quad [3]$$

(ii) Hence, by considering either the sum or product of the roots of this quadratic equation, or otherwise, determine the coordinates of B . [3]

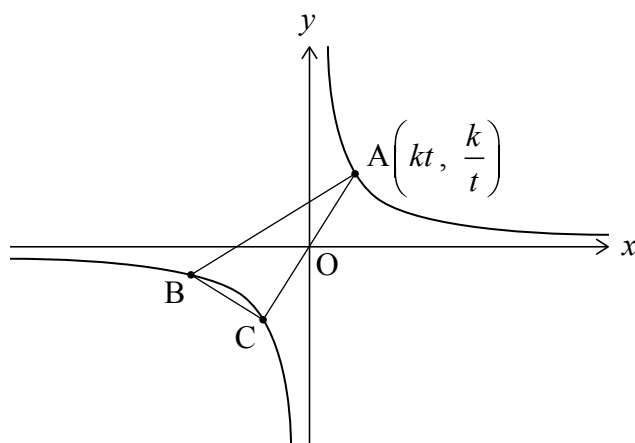
(This question continues on the following page)

(Question 1 continued)

From A , the line passing through the origin O intersects F again at point C .

Points A , B and C form triangle ABC as shown in the following diagram.

diagram not to scale



- (f) Prove that $\hat{BCA}$ is a right angle.

[6]